

BRANDT KAUFMANN

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Machine Learning Engineer with 3 years of computer-vision experience across research and production

Education

University of San Francisco

M.S. in Data Science

B.S. in Data Science

San Francisco, CA

June 2025 – June 2026

August 2022 – May 2025

Experience

Machine Learning Engineer

San Francisco, CA

XDR Radiology

September 2025 – Present

- Built a production-ready multi-model Azure ML online endpoint hosting 7 computer-vision models (YOLO object detection, classification, and segmentation) that flag anatomical structures and imaging-technique errors in dental X-rays; projected to cut radiologist and clinician image-inspection labor by 90%
- Containerized models with Docker and inference through a FastAPI service with Azure Managed Online Endpoints
- Built end-to-end MLOps pipeline: MLflow experiment tracking, model registry, and CI/CD with automated traffic routing and one-command deployment
- Trained EfficientNetV2, Attention-UNet, and YOLO models for X-ray classification, segmentation, and object detection
- Lifted classification model accuracy from 85% to 99% via custom image augmentation, bagging, and hp tuning
- Developed an in-house annotation tool for classification and bounding-box (object detection) datasets: a Python app that serves a browser UI for users to add, edit, and delete labels, syncing changes directly to the live Azure dataset to accelerate creation of new training data

Graduate Research Scientist

San Francisco, CA

University of San Francisco

June 2024 – August 2025

- Collected, trained, and fine-tuned CNNs for strong gravitational lens discovery in the DR10 Legacy Survey
- Distributed training over 4 GPUs at Lawrence Berkeley National Laboratory, reducing wall-clock training time by $3.7\times$
- Achieved AUC-ROC 99.997% on $\sim 140K$ galaxies; ran inference on 40M+ images (~ 5 TB) using a meta-learner ensemble
- Discovered 800+ new strong lens candidates via transfer learning; co-authored work accepted by *The Astrophysical Journal* (preprint on arXiv)

Research Intern

San Francisco, CA

Big Data REU (UMBC)

May 2023 – September 2023

- Applied RNNs to enhance gamma-ray images for radiotherapy; migrated to PyTorch and improved accuracy by $\sim 10\%$ with hyperparameter tuning
- Presented research at the 2023 International Conference on Machine Learning and Applications

Publications

- CNNs: Strong Lens Discoveries in DESI Legacy Imaging Surveys DR10 with Two Deep Learning Architectures
- Image Enhancement: Accelerating Real-Time Imaging for Radiotherapy

Projects

Snake DQN Agent — PyTorch/Pygame Reinforcement Learning

July 2026

- Designed and trained a Deep Q-Network to play Snake end-to-end, engineering a multi-channel grid state representation alongside hand-crafted scalar features, with epsilon-greedy exploration, experience replay, and short/long-memory steps
- Diagnosed a recurring failure mode via manual play analysis where the agent trapped itself in dead-end pockets formed by its own body; designed and implemented a custom BFS-based reachable-space algorithm to quantify accessible board area per candidate move and folded it into both the state features and reward shaping
- Built a checkpointing system separating protected `best.pth` weights from resumable `latest.pth` weights, with CLI flags for safely continuing training runs across sessions without risking regression

Technical Skills

Languages: Python (OpenCV, NumPy, Matplotlib, Scikit-learn), SQL (MySQL, PostgreSQL), NoSQL (MongoDB), PySpark
Computer Vision & ML: Computer Vision, CNNs, Object Detection (YOLO), Image Classification, Segmentation, Transfer Learning, Fine Tuning, Distributed Training, High Performance Computing, TensorFlow, PyTorch, AI, Data Structures
MLOps & Cloud: MLflow, Model Registry, Docker, FastAPI, CI/CD, Hyperparameter Tuning, Azure Machine Learning, Managed Online Endpoints, Google Cloud Platform(GCP), Airflow
Tools: Git, GitHub, Azure DevOps, Linux/Unix,